

LESSON 314 (2.0 GROUND)

Lesson Objective

During this ground lesson, the student will be introduced to VOR/VORTAC, DME, intersection, NDB, and localizer holding procedures.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Instrument Flying Handbook. ([FAA-H-8083-15B](#))
 - Chapter 10 - IFR Flight:
 - Section 10-10.
- Aeronautical Information Manual. ([AIM](#))
 - Section 5-3-8: holding.

Lesson Content

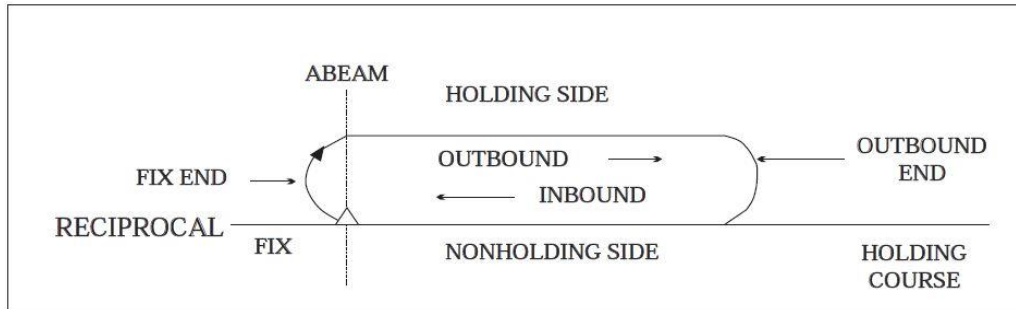
Holding Procedures

- Purpose of holds:
 - Controllers assign holds for a variety of reasons, including deteriorating weather or high traffic volume. Holding might also be required following a missed approach. Since flying outside the area set aside for a holding pattern could lead to an encounter with terrain or other aircraft, aircrews need to understand the size of the protected airspace that a holding pattern provides.

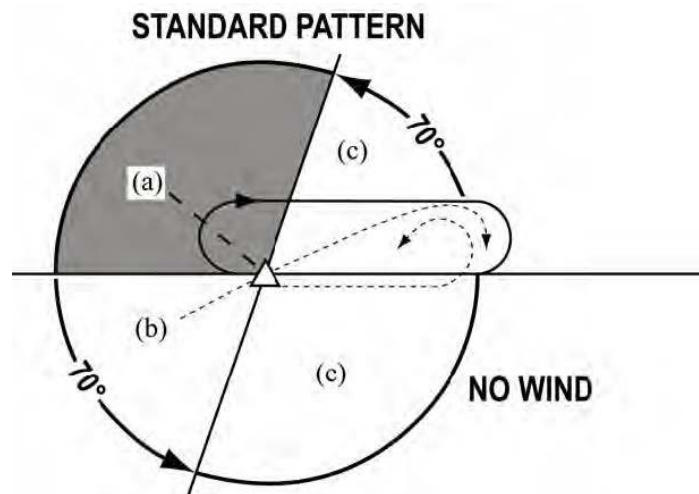
ATC clearances: AIM 1.8

- If not charted on the low enroute chart or on an approach plate, pilots can expect or request a clearance from a controller. Every holding clearance will have the following information.
 - *Direction of holding*
 - Every clearance will mention one of the eight cardinal direction eg. N,NE,E,SE,S etc.. When giving your student instructions, ensure that the radial or course matches that cardinal direction.
 - *Holding fix*
 - This can be a GPS fix, DME fix on a radial, an intersection between two radials, a fix on a localizer, or a VOR. Along with the fix, a radial, course, bearing, airway, or route on which the aircraft is to hold will be specified. (More on this below).
 - *Duration*
 - Based on length in GPS DME or time. Based on what is charted, or as requested by the pilot or ATC.

- *Direction of turn*
 - If no direction is mentioned in the clearance, it is understood to be right turns (standard). Left turns will be specified by the controller or upon the pilot's request.
- *Expect further clearance (EFC) time*
 - Time to expect further clearance, and any pertinent additional delay information.
- Elements of a hold AIM 1.9



Holding pattern entries: AIM 1.9.2.2



After determining the entry, use the 6 Ts

- **Direct Entry - (c)**
 - **Turn** - upon crossing fix, turn to the outbound heading.
 - **Talk** - advise ATC established in the hold.
 - **Time** - start the timer when the fix is crossed or wings level, whichever happens **last**.
 - **Twist** - twist to the inbound course.
 - **Throttle** - adjust throttle for 90 kts.
 - **Track** - after one minute or the holding distance, turn inbound and track the inbound course.

- **Parallel Entry - (a)**
 - Turn - upon crossing fix, turn to the outbound heading.
 - Time - start a timer for one minute.
 - Twist - twist to the inbound course.
 - Track - after 1 minute, make a 225° turn to track the inbound course.
 - If standard hold, it will be a left 225° turn.
 - If non-standard hold, it will be a right 225° turn.
 - Throttle - adjust throttle for 90 kts.
 - Talk - advise ATC established in the hold.

- **Teardrop Entry - (b)**
 - Turn - After crossing the fix, offset the outbound heading by 30 degrees.
 - Standard holds: add 30° to outbound heading.
 - Non-Standard holds: subtract 30° from outbound heading.
 - Remember: **LARS** (Left add, Right subtract).
 - Time - start a timer for 1 minute.
 - Twist - twist to the inbound course.
 - Track - after 1 minute, make a standard rate turn to join the inbound leg.
 - If standard hold, it will be a right turn.
 - If non-standard hold, it will be a left turn.
 - Throttle - adjust throttle for 90 kts.
 - Talk - advise ATC established in the hold.

Maintaining the holding pattern

- Wind correction:
 - Once established inbound for the first time, note the difference between the current heading and the ground track on the PFD.
 - This difference x 3 should be applied into the direction of the wind to the outbound heading.
 - Example: While established inbound, the difference between the current heading and ground track is 5°. Apply a 15° correction into the wind to the outbound heading.

- Time correction:
 - Time each full inbound leg from wings level until reaching the fix.
 - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
 - Note the difference between this time and one minute.
 - Add or subtract this difference from one minute to time each outbound leg.
 - Example: On the first full inbound leg, the time from wings level to reaching the fix

is 0:50. The outbound heading should now be timed to be 1:10.

- Continue utilizing the 6Ts and adjusting wind and time corrections throughout the hold.
 - No need to continue advising ATC of being established in the hold after initial contact is made.

Leaving the holding pattern

- We must advise ATC prior to leaving the holding pattern. ATC may assign specific instructions on how to exit.

Holding speeds and altitudes

- Maximum holding airspeed:
 - Below 6,000 ft. MSL: 200 KIAS.
 - 6,001 ft. MSL to 14,000 ft. MSL: 230 KIAS.
 - Above 14,000 ft. MSL: 265 KIAS.
- Standard holding leg length (inbound leg):
 - At or below 14,000 feet MSL: 1:00
 - Above 14,000 feet MSL: 1:30
 - Timing begins over/abeam the fix, whichever occurs later. If the abeam position cannot be determined, start timing when the turn to the outbound leg is completed.

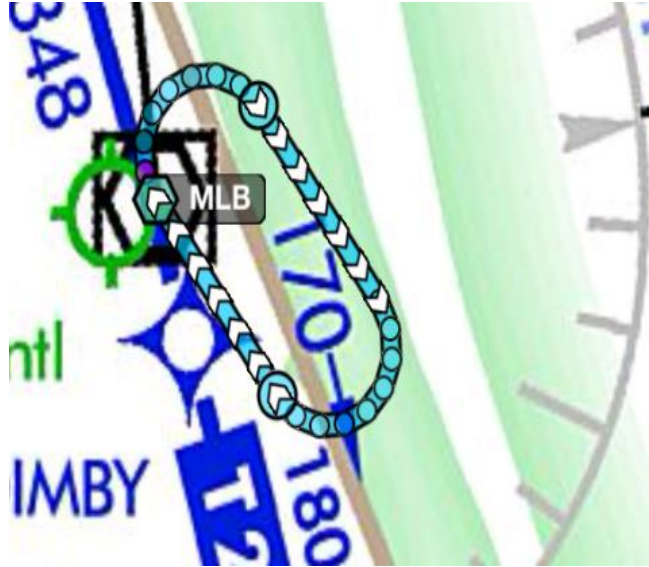
VOR/VORTAC holding patterns

- Holding instructions are given as radials.
 - The given radial is the outbound heading. Use this to determine the entry using the pencil method.
 - Twist to the reciprocal of this given radial to set the inbound course.
- The abeam point on the outbound leg is determined by a flag flip.
- Use caution as sensitivity increases the closer you are to the VOR/station.

“TECH FLIGHT 636, Hold SE of the MLB VOR on the 150 radial, EFC 1250Z”

→ Outbound heading: 150.

→ Inbound course: twist to 330.



Airway/DME holding patterns

- Emphasize the importance of the cardinal direction.
- The fix is determined by a DME distance; therefore, there will be **no flag flip**.
- The examples below are the same instruction with the cardinal direction changed as you can hold two sides of a DME fix.
- To eliminate confusion, the entry for DME holds can either be direct or parallel. Teardrop can still be utilized.
 - If direct, there will be no need to change the CDI. The outbound heading will be the reciprocal of the current course.
 - If parallel, the CDI will need to be flipped around to the reciprocal of the current course. The outbound heading is the current heading.
- To identify a DME fix on Foreflight, insert the VOR/radial/DME into the FPL tab when on the map page; for the example below, insert MLB/180/10.
- The abeam point on the outbound leg is determined by crossing assigned DME, there is **no flag flip**.

“TECH FLIGHT 636, Hold south of the 10 DME fix on the 180 radial from the MLB VOR, EFC 1409Z”

- The outbound heading is 180.
- To twist for the inbound leg, we set the 180 radial inbound (360).

Instrument Rating Course



"TECH FLIGHT 636, Hold north of the 10 DME fix on THE 180 radial from the MLB VOR, EFC 1510"

- Outbound heading: 360.
- Inbound course: twist to 180.



Intersection holding patterns

- This is achieved by using the intersection of two VOR radials as a fix.
- This can also be done by using a fix on a V-airway.
- For example: TPSTR can be identified using the GPS; For an intersection hold on a radial, we will identify this fix using the 16 DME from the MLB VOR on V347.
 - To twist the inbound leg, we will set the 189 radial.
- The abeam point on the outbound leg is determined by crossing assigned DME, there is **no flag flip**.

"TECH FLIGHT 636, Hold SW of TPSTR on V437, left turns, EFC 0611Z"

- Outbound heading: 189.
- Inbound course: twist to 009.

Instrument Rating Course

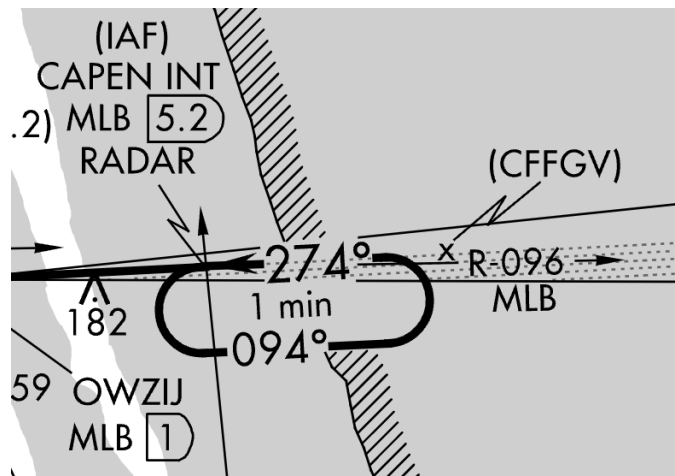


Localizer holding patterns

- Can hold on either side of a fix on a localizer, typically given for spacing due to traffic congestion.
- Fix can be identified using GPS or DME.
- Holds can be published or as given by ATC.

"Hold over CAPEN as published."

- Outbound heading is 094.
- Inbound course: on the localizer, twist to 094.



GPS holding patterns

- GPS fix can be input manually into G430 OR by navigating to the NRST pg on the G430 and scrolling to the nearest intersections.
- Remember to **press OBS** after reaching the fix. This will allow you to twist the CDI and set the

holding course.

- When giving instructions, we use either course or bearing; cardinal direction is key to identify.

“TECH FLIGHT 636, Hold NE of VALKA on the 040 course, left turns, EFC 1128Z”

- Outbound heading: 040.
- Inbound course: press OBS, then twist to 220.



- Published GPS holds as part of an approach
 - [**Note:** Revisit this again during the approaches ground/sim]
 - When doing an approach with a published hold, ATC can instruct you to hold as published or you request a publish hold/procedure turn.
 - To hold as published
 - PROC → Select approach
 - Make sure correct airport identifier is listed before selecting an approach
 - In this example, we'll choose the RNAV 9R @ KMLB
 - Select "SUROY" as your IAF, not vectors
 - The G430 will prompt if you would like to hold at SUROY, press yes
 - Around 0.3 NM away, the G430 will indicate the holding entry and heading to fly.
 - For a teardrop entry
 - If the length of the hold was distance (4 DME) and not time (eg. 1 min), we cannot offset the outbound heading by 30 degrees and fly it for 4 DME as this would lead us to the unprotected side of a hold. There are two methods:
 - Method 1
 - 1) Use the steps mentioned above in "To hold as published".
 - 2) The G430 will indicate a teardrop entry.

